The University of the South Pacific

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School of Information Technology, Engineering, Mathematics and Physics

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CS211: Computer Organisation

Test 2 – Semester 1, 2022

Time Allowed: 50 mins

Total Marks: 30 (15% of final grade)

Name: _____ **Student ID:** _____

INSTRUCTIONS:

- There are 7 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

1. Explain how a branch prediction technique improves the performance of a machine at the microarchitecture level. (3 marks)
   Ans:
   Branch prediction is very important to the performance of a deeply pipelined processor. Branch prediction **enables the processor to begin executing instructions long before the branch outcome is certain**. Branch delay is the penalty that is incurred in the absence of a correct prediction.

2. Explain the effects of having fixed length instructions on memory. (3 marks)
   ANS:
   . It wastes memory which requires padding the shorter instructions so as to make it of fixed length.
   • The program needs more memory at the time of execution of the instruction and also requires larger time to load from the secondary memory.

3. Explain how a D-flip-flops behaviour is different from D Latch behaviour. (3 marks)

   ANS: In a D flip-flop, the memory value changes only at that instant that the input becomes 1. In a latch, however, the memory value continues adopting any values given as long as its input is 1. (In a D flip-flop, if the input changes while _ remains 1, the remembered value doesn't

   change. In a latch, however, the input changing while is 1 results in a change to the remembered value.)

4. Assume the following instructions are available for 0-address (stack-based) and 1-address machines: ADD, SUB, MUL, DIV, LOAD, STORE, PUSH, POP. Memory addresses like A, B, C, etc. are used as operands. Each instruction is made up of 24 bits operand and 8 bits of opcode.

   a. Write instructions for a 0-address (stack) machine to compute X = (A-B)/(C+D/E). (3 marks)

      ```
      PUSH  A
      PUSH  B
      SUB
      PUSH  D
      PUSH  E
      DIV
      PUSH  C
      ADD
      DIV
      POP   X
      ```

   b. How many bits does 0-address machine need to compute X? (1 mark)

      224

   c. Write instructions for a 1-address machine to compute X = (A-B)/(C+D/E). (3 marks)

      ```
      LOAD  D
      DIV   E
      ADD   C
      STORE T
      LOAD  A
      SUB   B
      DIV   T
      STORE X
      ```

   d. How many bits does 1-address machine need to compute X? (1 mark)

      256

5. Design an expanding opcode to allow the following to be encoded in a 22-bit instruction with each address being 10bits. (4 marks)

3 two-address instructions
24 one-address instructions
15 zero-address instructions

ANS:

00	AAAAAAAAAA	BBBBBBBBBB
01	AAAAAAAAAA	BBBBBBBBBB
10	AAAAAAAAAA	BBBBBBBBBB
11	0000000000	BBBBBBBBBB
	.	
	.	
11	0000010111	BBBBBBBBBB
11	0000011000	0000000000
11	000001100	0000001110

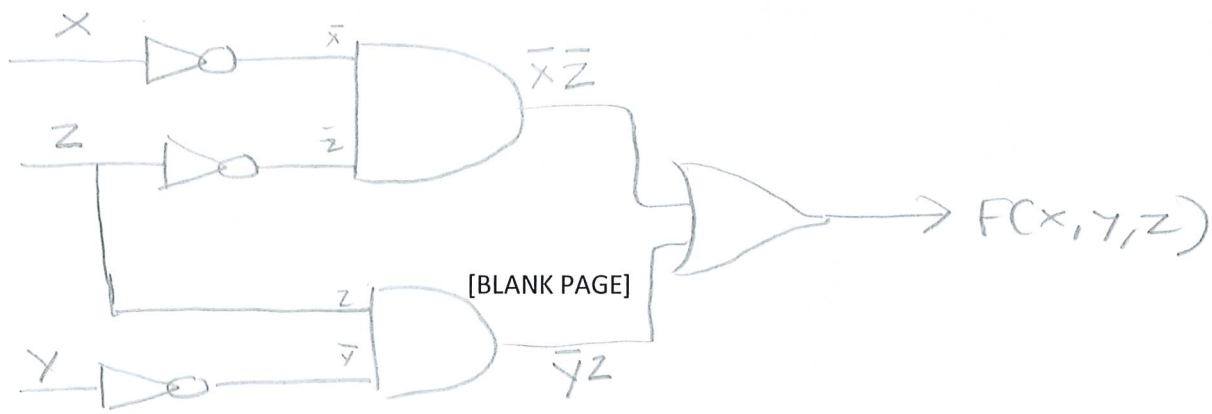
6. A function F is defined in the truth below below:

x	y	z	F(x,y,z)
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

a. Write the disjunctive normal form (DNF) for the above function F. (2 marks)
b. Simply the above DNF expression in its simplest form. (3 marks)
c. Draw the circuit diagram for the simplest form in part b) above. (2 marks)
d. Use the above truth table to draw the three variable multiplexer chip for the function F. (2 marks)

ANS:

a) $F(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}z$
b) $\bar{x}\bar{Z} + \bar{y}z$



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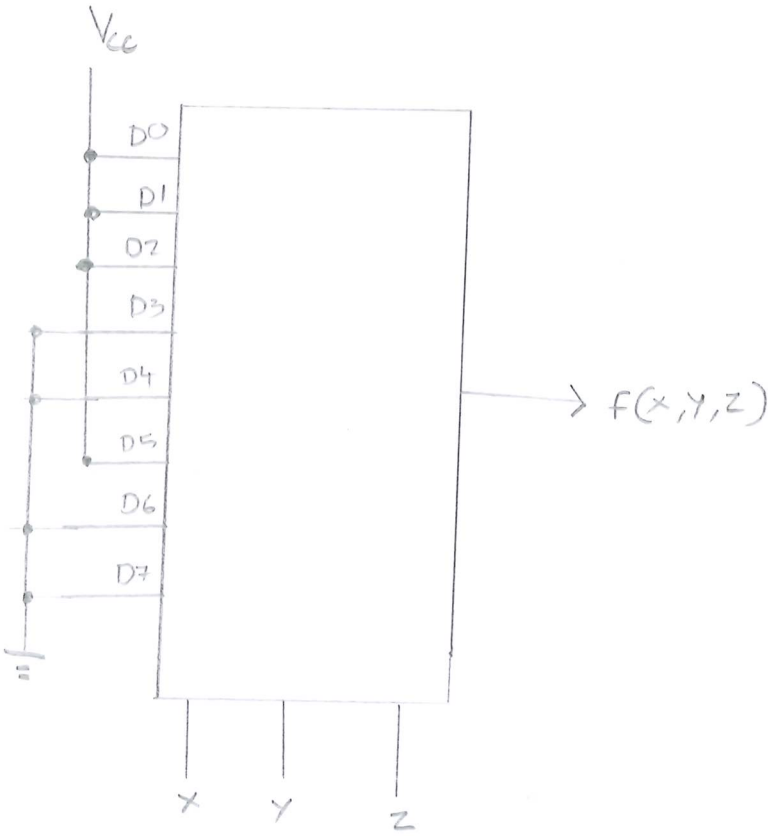
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